

The Non-Poisson Λ -Limit for k -Subsets

Exact matrix formula, proof, and exhaustive small-rank checks

Computational verification note

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Abstract

For fixed k , we prove the joint limit of all power traces of the permutation representation of S_n on k -subsets. The limit is an explicit functional of independent $Z_j \sim \text{Poisson}(1/j)$ variables and is non-Poisson for $k \geq 2$. An independent program constructs 1,584 representation matrices in dimensions 6, 10, 15, 20 and performs 6,336 exact power-trace comparisons.

1 The representation and exact formula

Let $\Omega_{n,k} = \binom{[n]}{k}$ and let $\rho_{n,k}$ permute its standard basis. Since the trace of a permutation matrix counts fixed basis vectors,

$$p_r(X_\rho(\sigma)) = \text{Tr}(\rho(\sigma)^r) = |\text{Fix}_{\Omega_{n,k}}(\sigma^r)|.$$

Write $A_j(\sigma)$ for the number of j -cycles of σ . A j -cycle splits under the r th power into $d = \gcd(j, r)$ cycles of length j/d . An invariant subset independently includes or excludes each resulting cycle, so

$$F_{k,r}(\sigma) = [u^k] \prod_{j=1}^{kr} \left(1 + u^{j/\gcd(j,r)}\right)^{\gcd(j,r)A_j(\sigma)}. \quad (1)$$

The cutoff is valid: if $j > kr$, then $j/\gcd(j, r) > k$.

For a partition λ , direct averaging therefore gives

$$\text{SMG}_{n,k} = \sum_{\lambda} \frac{\mathbb{E}_{S_n} [\prod_a F_{k,\lambda_a}(\sigma)]}{z_{\lambda}} p_{\lambda}. \quad (2)$$

2 Coefficientwise limit

Fix a finite set of exponents. Every variable in (1) is a polynomial in finitely many cycle counts. We use the following elementary factorial-moment lemma. For fixed nonnegative integers $r_1, \dots, r_R$,

$$\mathbb{E} \prod_{j=1}^R (A_j)_{r_j} = \prod_{j=1}^R j^{-r_j} \quad \text{whenever} \quad n \geq \sum_{j=1}^R j r_j.$$

Indeed, the falling-factorial product counts ordered selections of pairwise disjoint distinguished cycles, with r_j cycles of length j . Choosing their supports and cyclic orders and then permuting the remaining labels gives $n! \prod_j j^{-r_j} / n!$. These are exactly the joint factorial moments of independent $Z_j \sim \text{Poisson}(1/j)$. Hence the finite vector of cycle counts converges jointly, and every factorial moment required by a fixed coefficient is eventually equal to its limiting value. Thus

$$(A_1, \dots, A_R) \implies (Z_1, \dots, Z_R), \quad Z_j \text{ independent, } Z_j \sim \text{Poisson}(1/j).$$

Thus

$$F_{k,r}(\sigma) \implies F_{k,r}^{(\infty)} := [u^k] \prod_{j=1}^{kr} (1 + u^{j/\gcd(j,r)})^{\gcd(j,r)Z_j},$$

jointly for every finite collection of r 's. Moment stabilization permits expectations in (2) to pass to the limit, proving the coefficientwise Λ -limit

$$\text{SMG}_{\infty}^{(k)} = \sum_{\lambda} \frac{\mathbb{E}[\prod_a F_{k,\lambda_a}^{(\infty)}]}{z_{\lambda}} p_{\lambda}.$$

For $r = 1$, $F_{k,1}^{(\infty)}$ has mean one. The action has rank $k + 1$ once $n \geq 2k$, hence its limiting second moment is $k + 1$ and its variance is k . A Poisson variable of mean one has variance one; consequently the limit is non-Poisson for $k \geq 2$. For example

$$F_{2,1}^{(\infty)} = \binom{Z_1}{2} + Z_2.$$

3 Exact finite formula checks

For each $(n, k) = (4, 2), (5, 2), (6, 2), (6, 3)$, the verifier enumerates all $n!$ permutations, induces their action on lexicographically ordered k -subsets, and materializes the corresponding zero-one matrix. Distinct byte keys prove that the constructed matrices form a faithful copy of S_n . It then compares the integer matrix traces $\text{Tr}(M^r)$, $1 \leq r \leq 4$, with (1). Six joint products indexed by $(1), (2), (3), (1, 1), (2, 1), (1, 1, 1)$ are averaged independently on both sides.

n	k	dimension	matrices	trace checks	$(\mathbb{E}F, \text{Var } F)$
4	2	6	24	96	(1, 2)
5	2	10	120	480	(1, 2)
6	2	15	720	2880	(1, 2)
6	3	20	720	2880	(1, 3)

All 6,336 trace comparisons and all 24 joint-average comparisons pass exactly. These exhaustive matrices verify the finite fixed-point identity, not the asymptotic convergence; convergence is proved by the factorial-moment lemma above.

4 Reproduction and reference

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.venv/bin/python for_this_guy/limiting_lambda_claims/
  symmetric_k_subsets/verification.py
.venv/bin/marimo edit for_this_guy/limiting_lambda_claims/
  symmetric_k_subsets/notebook.py
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Join each wrapped command onto one line. For the fixed-point and moment background, see P. Diaconis, J. Fulman, and R. Guralnick, *On fixed points of permutations*, <https://arxiv.org/abs/0708.2643>.